In Criteria A, we understood the problem, researched information and learned about users. Now it's time to become designers

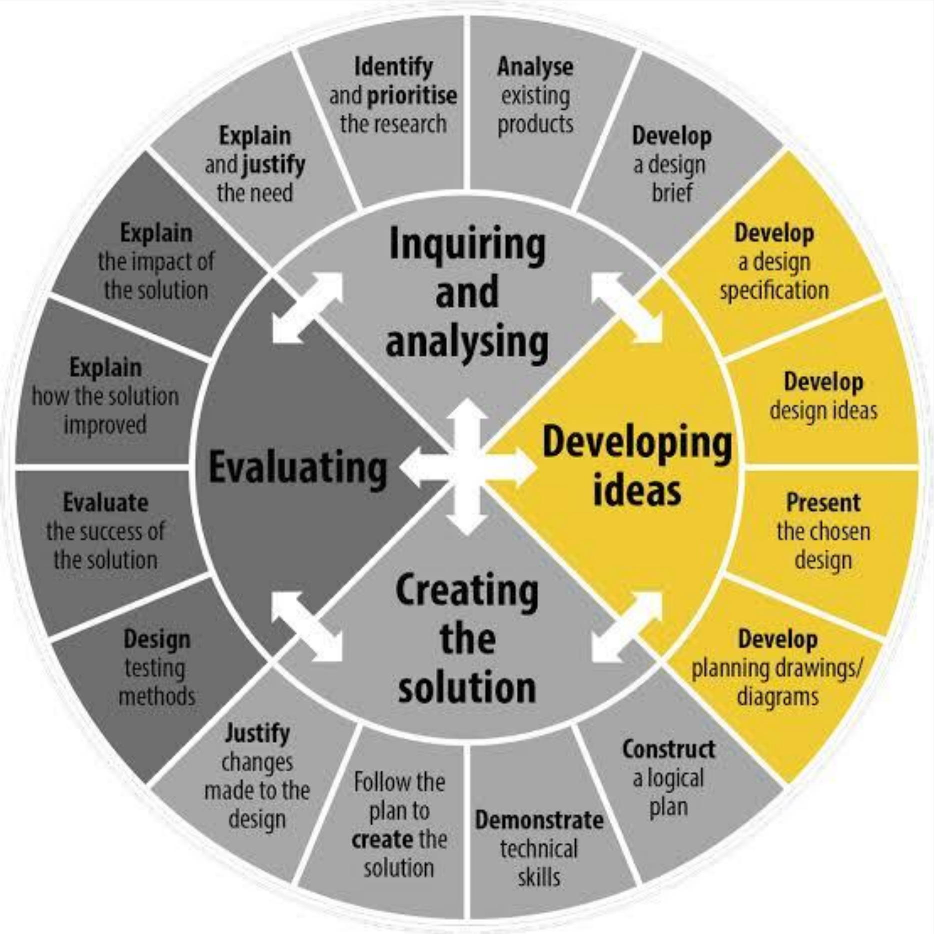
- Can we start building immediately?
- "Designers first think of ideas. They compare ideas and choose the best one before they start creating."

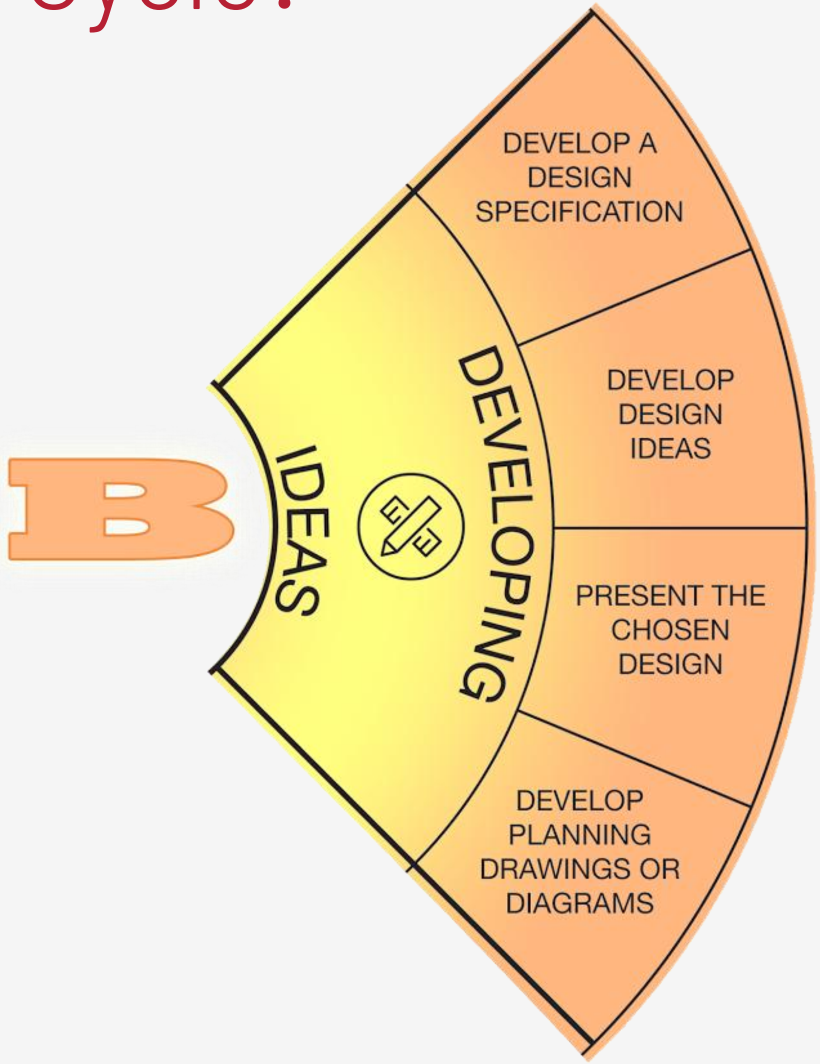
Research
↓
Ideas
↓
Create

Criteria B
Developing Ideas

Design the actual solution

Where Are We in the Design Cycle?





Strand i: develop a list of success criteria for the solution

Command Term: Develop

Meaning

- To improve or expand an idea by adding more detail.

Example

- Bad idea : "My game should be good."
- Developed idea: "My game should have clear instructions, colourful sprites, easy controls and immediate feedback."

Develop = Add more useful details.

School Bag Specifications

- ✓ Fits A4 books
- ✓ Waterproof
- ✓ Comfortable straps
- ✓ Strong zip
- ✓ Lightweight

Every specification is:
Clear ✓
Specific ✓
Easy to check ✓

Our game specifications should also be clear, specific and easy to check.

Success Criteria

What are Success Criteria?

- Success Criteria are the things your solution **must have** to be successful.
- They act like a checklist that guides your work.
- Think of them as **"Must Have" features** for your project.

Why Do Designers Write Success Criteria?

Success criteria help designers:

✓ Stay focused

✓ Know what the solution should include

✓ Check whether the solution is successful

Without success criteria, it is difficult to know if a design meets the user's needs.

Everyday Example

School Bag

Imagine you are designing a school bag.

Your success criteria might be:

✓ Fits A4 books

✓ Waterproof

✓ Comfortable straps

✓ Strong zip

✓ Lightweight

- If your bag meets all these points, it is a successful design.

Summative Task

Criteria B: Developing Ideas

Before creating your interactive digital solution, you will generate ideas, develop success criteria, select the best design and create a plan for making your solution.

Strand i: Develop a list of success criteria for the solution

Activity: Success Criteria Development

Thinking Skills – Critical Thinking – Propose and evaluate a variety of solutions

Based on your research and design brief, develop a list of success criteria that your solution should meet. Your success criteria should help you judge whether your final solution is successful.

Examples:

- The game should be easy to understand and play.
- The user should be able to interact with the game.

Create at least **5 success criteria** for your solution.

NO.	Success criteria	Why is it important?
1		
2		
3		
4		
5		

Strand ii: Present feasible design ideas, which can be correctly interpreted by others.

Command Terms

Present: Offer information clearly so that others can understand it.

Feasible: Possible to create using the available time, knowledge and tools.

Example:

❌ **Not Feasible**
A game with 100 levels, online multiplayer, and AI characters.
(Too difficult for this project.)

✅ **Feasible**
A Guess the Number Game made using PictoBlox.
(It can be completed using the coding blocks you have learned.)

What Are Design Ideas?

Before coding, designers first **plan** their ideas.

A design idea shows:

- What the user will see
- What the user will do
- How the program will respond

It is like a blueprint for your project.

Why Do Designers Draw Their Ideas?

Imagine asking someone to build your game.

- If you only say:

  "Make a guessing game."

- Everyone will imagine something different.

Instead, designers **draw and explain** their ideas so everyone understands the same thing.

What Makes a Good Design Idea?

A good design idea should show:

✓ Screen layout

✓ Buttons or sprites

✓ User input

✓ Program output

✓ Interaction

✓ Labels

Components of a Design Idea

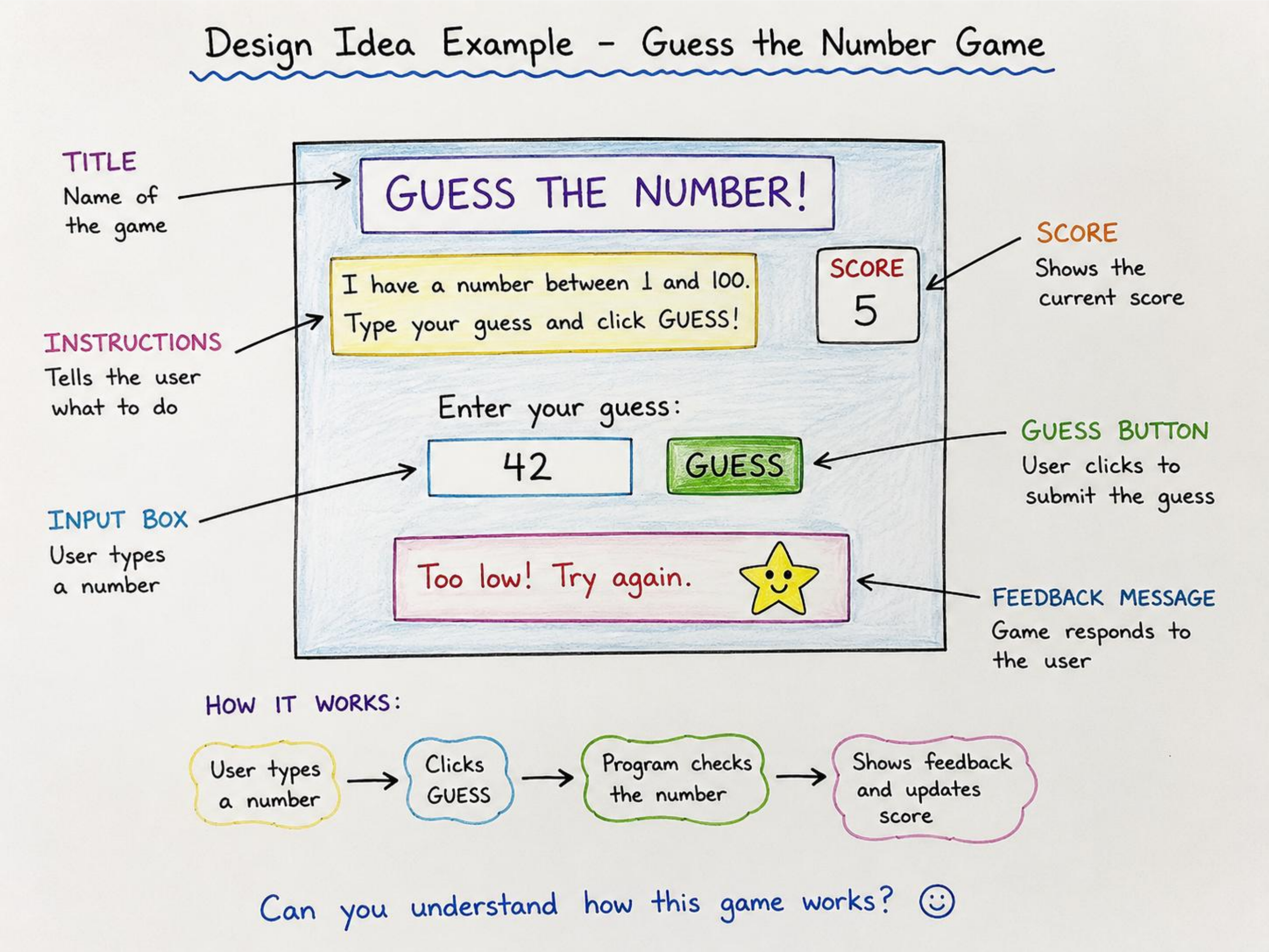
Your design idea should include:

🖥 Screen Layout
Where are the sprites?
Where are the buttons?
Where is the score?

⌨ Input
What will the user do?
Click?
Type?
Press a key?

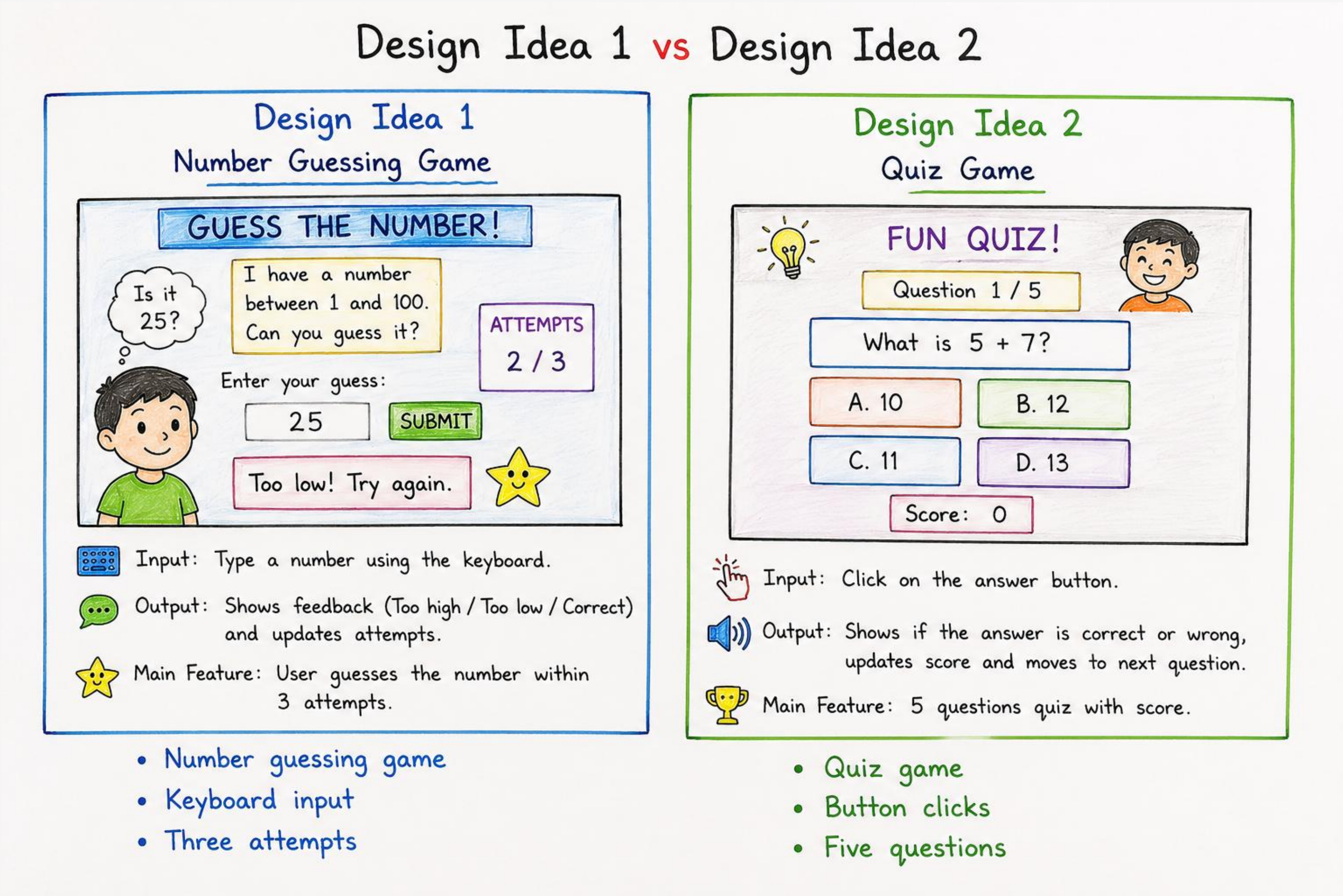
💻 Output
What will the program show?
Message?
Score?
Animation?
Sound?

Example Design Idea



Like this you need to Include labels such as:

- Title
- Instructions
- Input box
- Guess button
- Feedback message
- Score



Which one would you choose?
Why?

Interaction Happens Here

User
↓
Types a Number
↓
Program Checks the Answer
↓
Shows "Correct!" or "Try Again"

"Every design idea should show **how the user and the program communicate**. This interaction is the most important part of your design."

Summative Task

Strand ii: Present feasible design ideas, which can be correctly interpreted by others

Activity: Generating Design Ideas

Thinking Skills – Creativity and Innovation – Use brainstorming and mind mapping to generate new ideas and inquiries

Develop and communicate different ideas for your interactive digital solution. Your ideas should clearly show:

- What the user will do
- How the program will respond
- The main interactive features

Present **two feasible design ideas**. For each idea include:

Design Idea 1

- Sketch / Screen Layout
- Description of interaction
- Inputs used
- Outputs provided

Design Idea 2

- Sketch / Screen Layout
- Description of interaction
- Inputs used
- Outputs provided

Design Idea 1	Design Idea 2
Description:	Description:

Strand iii: Present the chosen design

Present: Offer information clearly so that others can understand it.

Choose: Select the best option from different ideas.

Explain: Give reasons and details.

Do Designers Use Every Idea?

- Imagine you designed two games.
- Should you create both?

🤔 **No!**

Designers compare their ideas and **choose the one** that works best.

How Do Designers Choose?

Designers ask questions like:

✅ Which idea meets the user's needs?

✅ Which idea is easy to create?

✅ Which idea is more interactive?

✅ Which idea meets the success criteria?

The design with the most "Yes" answers is usually the best choice.

Let's Compare Two Ideas

Feature	Design Idea 1	Design Idea 2
User can interact	✅	✅
Easy to create	✅	❌
Gives feedback	✅	✅
Uses learned coding blocks	✅	❌

Evaluation Table Example

Criteria	Design 1	Design 2
Meets user needs	✅	✅
Easy to create	✅	❌
Interactive	✅	✅
Meets success criteria	✅	⚠️ Partly
Suitable for users	✅	✅

Which design would you select?

Why Did You Choose It?

- A good designer gives reasons.

❌ Weak Answer
"I like Design 1."

✅ Better Answer
"I chose Design 1 because it is easy to create, meets the success criteria, and allows users to interact with the program through keyboard input and immediate feedback."

Summative Task

Strand iii: Present the chosen design

Activity: Choosing the Best Design

Thinking Skills – Critical Thinking – Evaluate evidence and arguments

Review your ideas and select the design that best meets your success criteria.

Task

a) Identify your chosen design.
b) Explain why you selected this design.
c) Explain how it meets your success criteria.
d) Explain why it is suitable for the intended users.

Criteria	Design Idea 1	Design Idea 2
Meets user needs		
Easy to create		
Interactive		
Meets Success Criteria		
Suitable for intended users		

Chosen Design Justification

____

____

____

____

____

Strand iv: Create a planning drawing/diagram which outlines the main details for making the chosen solution.

Command Term

Create: Produce or make something new.

Planning Drawing / Diagram: A visual plan that shows how the solution will work before we create it.

Why Do Designers Plan?

- **Would you build a LEGO model without looking at the instructions?**
- **Would you bake a cake without following the recipe?**

Probably not!

Designers also make a **plan before they build.**

What is a Flowchart?

A **flowchart** is a diagram that shows the steps of a program from beginning to end.

It helps us understand:

✓ What happens first

✓ What happens next

✓ What decision the program makes

✓ What happens at the end

Symbol	Meaning
⬭ Oval	Start / Stop
□ Rectangle	Process (an action)
◇ Diamond	Decision (Yes / No)
▱ Parallelogram	Input / Output
→ Arrow	Direction of the program

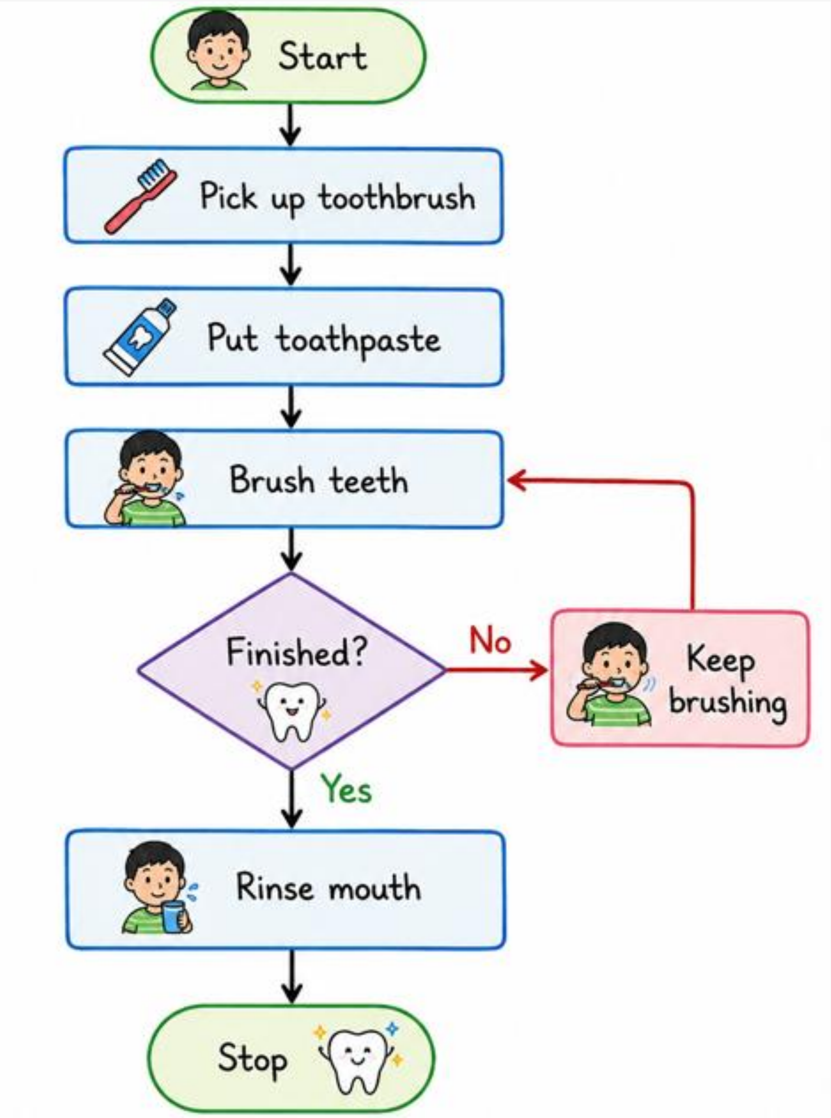
Before you draw

Your flowchart should include:

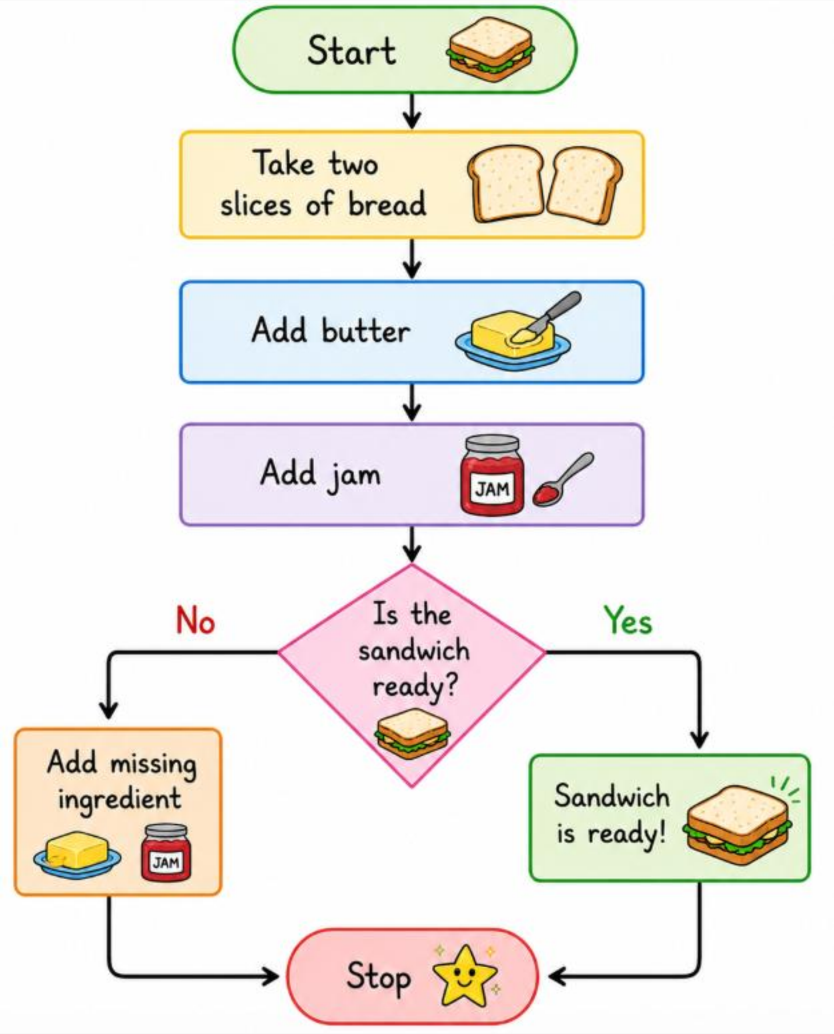
- ✅ Start
- ✅ Input
- ✅ Process
- ✅ Decision
- ✅ Output
- ✅ Stop

And arrows showing the order.

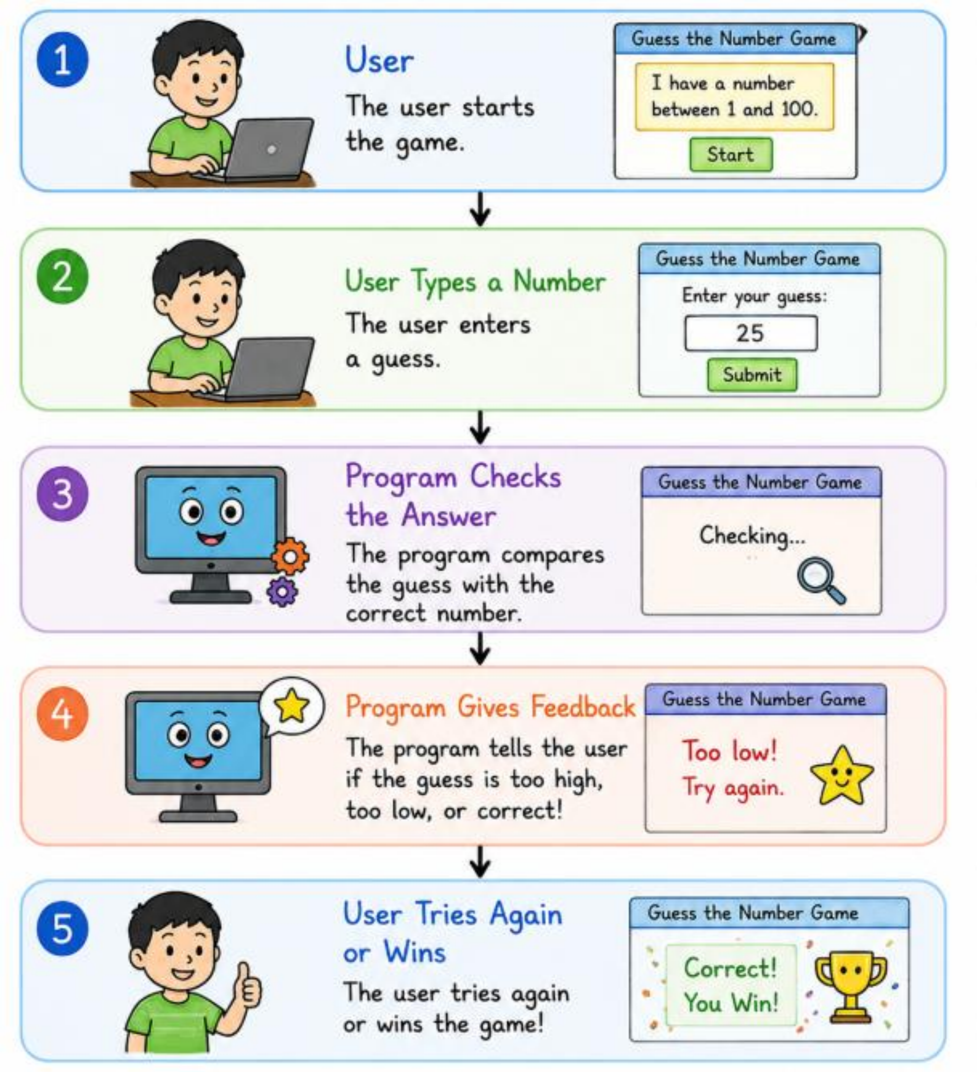
Example Flow chart: Brushing Your Teeth



Flowchart of Making a Sandwich



User Interaction- How they work together



Summative Task

Strand iv: Create a planning drawing/diagram which outlines the main details for making the chosen solution

Activity: Planning the Solution

Self-management Skills – Organization – Plan short and long-term assignments; meet deadlines

Create a detailed **flowchart** showing:

- Start
- Input
- Process
- Decision
- Output
- Stop

The flowchart should clearly show how the user interacts with the solution and how the program responds.

Flowchart for Our Guess the Number Game